



SWARM DRONE CHALLENGE

Regulations

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Ingolstadt



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Preface

Welcome to the Swarm Drone Challenge!

An evolution of the 2021 Deep Drone Challenge, the upcoming 2024 Swarm Drone Challenge will push the boundaries of drone technology as international teams present their creative and innovative algorithms. The focus on swarms opens new horizons for the fields of artificial intelligence and machine learning.

In the “Swarm Drone Challenge 2024”, participants develop solutions in which drones cooperate in a swarm and solve tasks together in competition. The “Swarm Drone Challenge 2024” is not only a stage for the competition, but also a platform for the exchange of ideas and techniques.

1 Competition Outlines

1.1 Playing Field

The playing field is a rectangular area with the following dimensions:

- Length: 36 meters
- Width: 18 meters
- Height: 6 meters

The playing field is divided into three zones:

- **Two Team Zones (red and blue):** The home base of each team, where the drones start and land.
- **One Neutral Zone in the middle:** A neutral territory for strategic maneuvers.

The field layout is illustrated in the figure below:

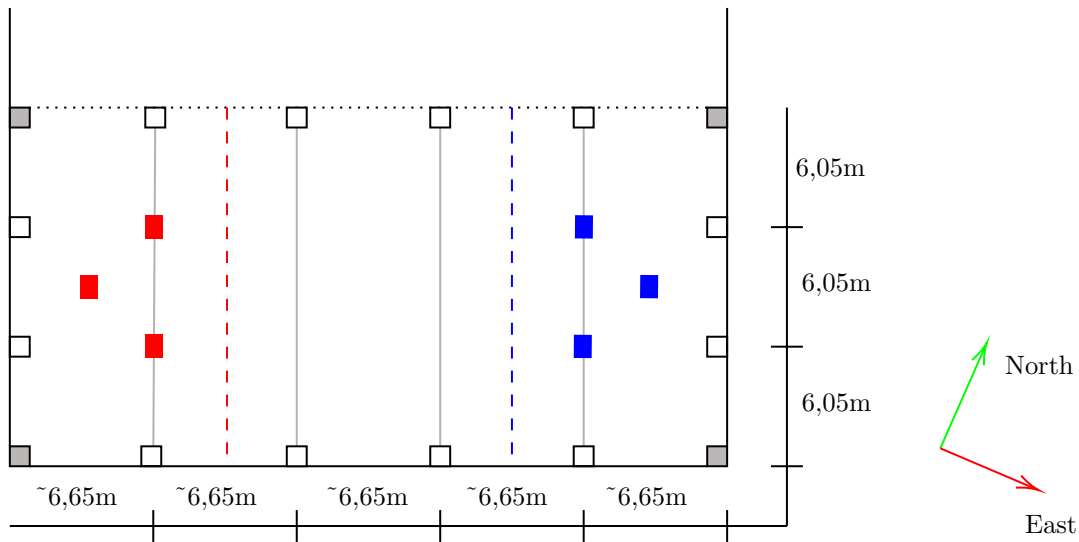


Figure 1: Field Dimensions

The sizes shown in the graphic are approximate values. The exact dimensions may vary slightly.

1.2 Target Objects

- **The goal:** Change the color of the target objects to your team's color (red or blue)!
- Each team zone has **target objects**, for a total of **6** in the playing field.
- These objects change their color based on their interaction with the drones:
 - **Default state:** Matches the color of the team zone.
 - **Triggered state:** Changes to the color of the drone that approaches and then flies away.

1.3 Gameplay

- **Teams:** 2-5 drones per team, ready to battle!
- **Start:** Each team starts in its designated zone with its drones powered up and ready to go.
- **Action:** Drones fly around the arena, triggering target objects to change color and score points for their team.
- **Scoring:** Each change of a target object's color earns your team 1 point.

- **Time Limit:** The game lasts for a maximum of 10 minutes. The team with the most points at the end wins!



1.4 Special Maneuver: Dominate the Arena!

- **Unlock the superpower:** When all 6 target objects are the same color (either red or blue), your team can attempt a special maneuver.
- **The challenge:** A drone from your team needs to fly from the enemy zone in your own zone. This drone or another drone has to land in your own zone afterwards.
- **Success!** If all targets remain the same color after this action, your team wins the game instantly!

1.5 Tournament

- Two teams compete against each other in each match.
- Pairings are determined by random draw.
- The winner of the tournament is the team that scores the most points in its match. Teams that win the match via a special maneuver are ranked at the top of the leaderboard.
- Each team plays only once.
- If the number of teams is odd, an opponent team is determined for the remaining team.
- Firstly, voluntary opponent teams are sought. If only one voluntary team is available, it is directly nominated. If there are multiple voluntary teams, the decision is made by random draw.
- If there are no voluntary teams, a random draw selects an opponent from all teams.
- The team selected as the opponent then has two game results, and the better result is considered.



1.6 Winning the Challenge

To win the Swarm Drone Challenge, a jury of experts will evaluate the teams' performance in the game as well as their technical approach and the presentation of their work. The winning team is determined by the jury.

1.7 Teams

- Up to 5 team members per team
- New team members are allowed to join but need to be registered with brigk.

1.8 Reimbursement

- Up to 1000€ for drone parts
- Max 5 orders
- Until the final, the hardware remains the property of brigkAIR and is provided to the teams as a permanent loan.

1.9 Penalty and disqualification

- Leaving the flight arena will be penalized with a deduction of 15 points.
- Starting the motors of the drones before the official start will be penalized with a deduction of 5 points



1.10 Intellectual property

- The IP stays at the teams
- The teams are encouraged to open source their solutions. This can have an effect on the jury voting
- The teams are required to prepare a presentation and explain their solution with technical details on the implementation.



1.11 Responsibilities

- Each team must have a spotter
- Teams are responsible to virtually validate their solution

2 Technical Regulations

2.1 Positioning

The challenge will take place in a hangar at the Manching Airport ($48^{\circ}43'21.7''\text{N}$ $11^{\circ}33'04.9''\text{E}$). To improve the state estimation in a potential gps denied environment, a camera based positioning system using ArUco marker can be used. Two ArUco marker are attached to each pillar in the hangar. The figure below illustrates the IDs of the marker that are attached to the pillars, where the lower ArUco marker is mounted at a height of 2m and the upper marker at 4m.

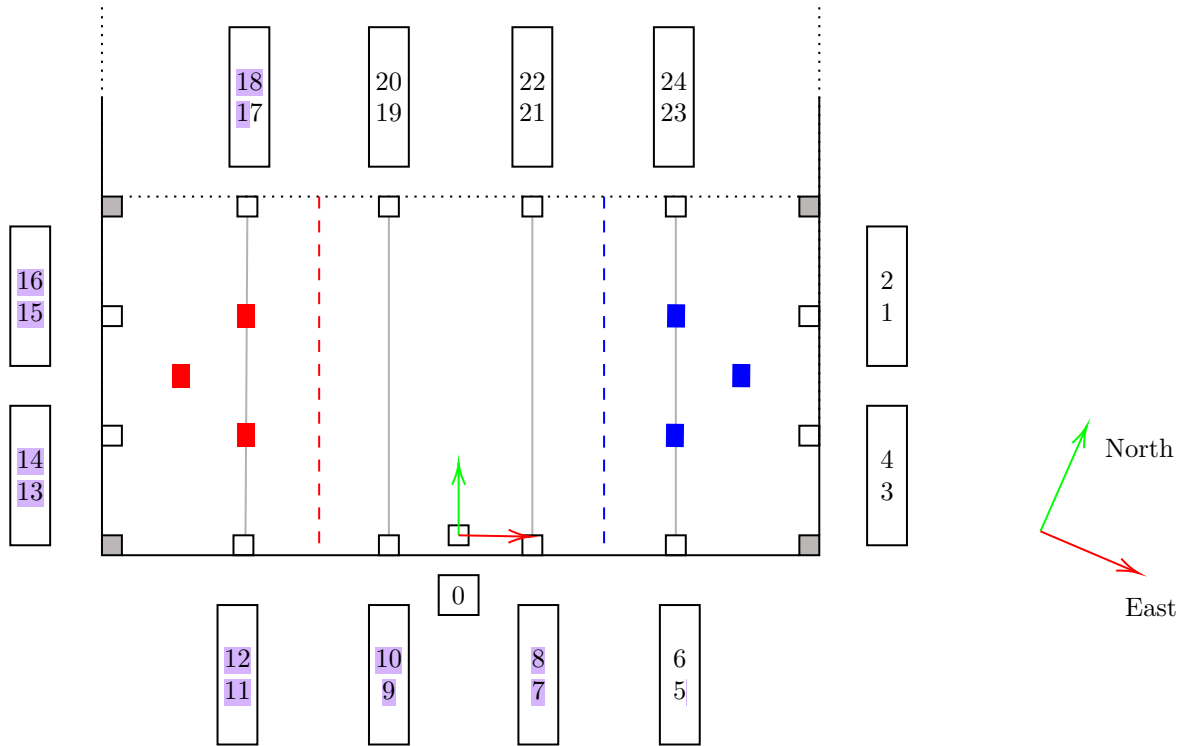


Figure 2: ArUco Marker ID's

None of the hangars walls are aligned to the north or the east direction. To simplify the positioning system an additional marker with the ID 0 is placed at the back wall in the center of the field. This 0 marker can be used to attach the local world coordinate system to it. The orientation of the marker is shown in figure 3

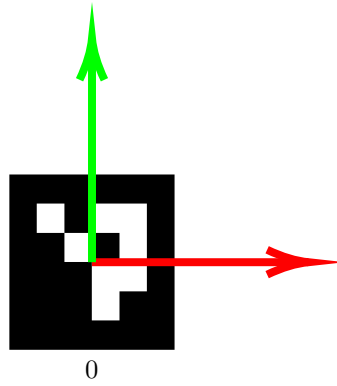


Figure 3: Zero marker for the local coordinate frame

The aruco marker are from the *DICT_{4x450}* dictionary and have a size of 0.5m x 0.5m. This information enables positioning in the arena with a view of a single marker

2.2 Collision Prevention

- Each drone is recommended to be equipped with a spherical protection case to protect the drone and other drones from damage

2.3 Emergency Shutoff and Kill Switch

- Each drone must be equipped with a physical or virtual kill switch.
- Activation of the kill switch results in immediate landing or motor Shutoff
- One button to land whole swarm

2.4 Size and weight limits

- Below 25kg per drone
- The drone swarm must consist of no more than 5 drones
- The drone swarm must consist of more that 2 drones.

2.5 Drone identification

- Each drone must be equipped with an RFID marker for identification by the target objects. The markers will be provided by brigkAIR at the event.

2.6 Ground Control Station

- It is not required that the computing must be performed *solely* onboard.
- The *behavior* of individual drones can be either controlled by a single agent, which is aware of the global state or by one agent per drone.

2.7 Arbiter / Judge

No technical solution is provided by the organizers that collects and/or distributes (real-time) information on positioning, scoring and thus acts as a source of truth or a referee.